

Matteo Facchini

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EDUCATION

Stony Brook University

May 2028

Bachelor of Science, Computer Science

GPA: 3.55 / 4.0

- Relevant Coursework: Analysis of Algorithms, Data Structures & Algorithms, Systems Programming, Discrete Mathematics, Object-Oriented Programming, Probability & Statistics, Linear Algebra

TECHNICAL SKILLS

- **Languages:** Python, C, Java, SQL, JavaScript, HTML, CSS
- **Frameworks/Libraries:** FastAPI, Pydantic, SQLAlchemy, React, Next.js, Streamlit
- **Databases/Technologies:** PostgreSQL, ChromaDB, OpenAI API, Docker
- **Developer Tools:** Git, GitHub, Linux, pytest, GitHub Actions

EXPERIENCE

Undergraduate Researcher — PACE Lab

September 2026 - Present

Stony Brook University | Advisor: Prof. Anshul Gandhi

- Investigating GPU resource sharing and workload co-location using NVIDIA MPS and CUDA Green Contexts, with a focus on performance isolation and resource partitioning.
- Benchmarking pairs of GPU workloads across different resource allocations and measuring latency and throughput to characterize performance behavior.
- Comparing GPU-sharing mechanisms and workload characteristics to study how resource allocation affects co-located workload performance.

Restaurant Waiter

June 2022 - September 2026

Stellina Restaurant, Oyster Bay NY

- Managed up to 18 tables at once in a fast-paced fine dining environment while coordinating with kitchen staff, bartenders, and servers under time constraints.

PROJECTS

RAG-Based Course Document Assistant (CourseLens AI) | [GitHub](#)

Python, FastAPI, OpenAI API, ChromaDB, Streamlit, PyMuPDF

- Built a RAG-based document assistant that processes course PDFs through text extraction, chunking, embeddings, ChromaDB vector search, and source-grounded LLM generation.
- Implemented 10+ FastAPI REST endpoints for course management, document ingestion, semantic retrieval, and AI question answering while preserving filename and page metadata for citations.
- Built a Streamlit interface and local persistence layer for uploading documents and interacting with indexed course materials.

Distributed CI/CD Testing Platform (ForgeCI) | [GitHub](#)

Python, FastAPI, PostgreSQL, SQLAlchemy, pytest, GitHub Actions

- Built a FastAPI service that processes GitHub push webhooks, authenticates requests using HMAC-SHA256 signatures, and extracts repository, commit SHA, and Git reference metadata.
- Integrated PostgreSQL connectivity using SQLAlchemy and psycopg with environment-based configuration for database credentials and application secrets.
- Developed 5 automated API and webhook-security tests with pytest and configured GitHub Actions to execute the test suite automatically on pushes and pull requests.

LEADERSHIP & INVOLVEMENT

Member, AI Community Club

April 2025 - Present

Stony Brook University